

Aprendizaje de Máquina

U2 Análisis de Clústeres

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Section 1

Divisive Algorithms

Divisive algorithms

The **divisive algorithms** follow the reverse strategy from that of the agglomerative schemes.

- The first clustering contains a single set, X .
- At the first step, we search for the **best possible partition** of X into two clusters.
- The straightforward method is to consider all possible $2^{N-1} - 1$ partitions of X into two sets and to select the optimum, according to a prespecified criterion.
- This procedure is then applied **iteratively** to each of the two sets produced in the previous stage.
- The final clustering consists of N clusters, each containing a single vector of X .

Divisive algorithm

The t -th clustering contains $t + 1$ clusters.

In the sequel C_{tj} will denote the j -th cluster of the t -th clustering R_t , $t = 0, \dots, N - 1$, $j = 1, \dots, t + 1$.

Let $g(C_i, C_j)$ be a dissimilarity function defined for all possible pairs of clusters.

The initial clustering R_0 contains only the set X , that is, $C_{01} = X$.

To determine the next clustering, we consider **all possible pairs** of clusters that form a partitioning of X .

Among them we choose the pair, denoted by (C_{11}, C_{12}) , that **maximizes** g .¹

¹It can be used a similarity function. In that case we should choose the pair of clusters that minimize g .

Divisive algorithm

These clusters form the next clustering R_1 , that is, $R_1 = \{C_{11}, C_{12}\}$.

At the next time step, we consider all possible pairs of clusters produced by C_{11} and we choose the one that maximizes g . The same procedure is repeated for C_{12} .

Assume now that from the two resulting pairs of clusters, the one originating from C_{11} gives the larger value of g . Let this pair be denoted by (C_{11}^1, C_{11}^2) .

Then the new clustering, R_2 consists of C_{11}^1 , C_{11}^2 , and C_{12} .

Relabeling these clusters as C_{21} , C_{22} , C_{23} , respectively, we have

$$R_2 = \{C_{21}, C_{22}, C_{23}\}$$

Carrying on in the same way, we form all subsequent clustering.

Generalized Divisive Scheme (GDS)

- Initialization
 - Choose $R_0 = \{X\}$ as the initial clustering.
 - $t = 0$
- Repeat
 - $t = t + 1$
 - For $i = 1$ to t
 - Among all possible pairs of clusters (C_r, C_s) that form a partition of $C_{t-1,i}$, find the pair $(C_{t-1,i}^1, C_{t-1,i}^2)$ that gives the maximum value for g .
 - Next i
 - From the t pairs defined in the previous step choose the one that maximizes g . Suppose that this is $(C_{t-1,j}^1, C_{t-1,j}^2)$.
 - The new clustering is

$$R_t = (R_{t-1} - \{C_{t-1,j}\}) \cup (C_{t-1,j}^1, C_{t-1,j}^2)$$

- Relabel the clusters of R_t .
- Until each vector lies in a single distinct cluster.

Efficiency

Different choices of g give rise to different algorithms.

One can easily observe that this divisive scheme is computationally very demanding, even for moderate values of N . This is its **main drawback**, compared with the agglomerative scheme.

Thus, if these schemes are to be of any use in practice, some further computational simplifications are required:

- One possibility is to make compromises and not search for all possible partitions of a cluster.

Split criterion

Let C_i be an already formed cluster. Our goal is to **split** it further, so that the two resulting clusters C_i^1 and C_i^2 , are as **dissimilar** as possible.

Initially, we start with

$$C_i^1 = \emptyset \quad \wedge \quad C_i^2 = C_i$$

Then, we identify the vector in C_i^2 whose **average dissimilarity** from the remaining vectors is maximum, and we move it to C_i^1 .

In the sequel, for each of the remaining $x \in C_i^2$, we compute its average dissimilarity with the vectors of C_i^1 :

$$g(x, C_i^1)$$

as well as its average dissimilarity with the rest of the vectors in C_i^2 :

$$g(x, C_i^2 - \{x\})$$

Split criterion

If for every $x \in C_i^2$, it holds

$$g(x, C_i^2 - \{x\}) < g(x, C_i^1)$$

then we stop.

Otherwise, we select the vector $x \in C_i^2$ for which the difference

$$D(x) = g(x, C_i^2 - \{x\}) - g(x, C_i^1)$$

is **maximum** (among the vectors of C_i^2 , x exhibits the maximum dissimilarity with $C_i^2 - \{x\}$ and the maximum similarity to C_i^1) and we move it to C_i^1 .

The procedure is repeated until the termination criterion is met.

Section 2

Choice of the best number of clusters

Choice of the best number of clusters

One important issue in hierarchical algorithms is determining the best clustering within a given hierarchy. This is equivalent to identifying the **number of clusters** that **best fits** the data.

Lifetime of a cluster

An intuitive approach is to search in the proximity dendrogram for clusters that have a **lifetime**.

The lifetime of a cluster is defined as the absolute value of the difference between the **proximity level** at which it is created and the proximity level at which it is absorbed into a larger cluster.

However, **human subjectivity** is required to reach conclusions.

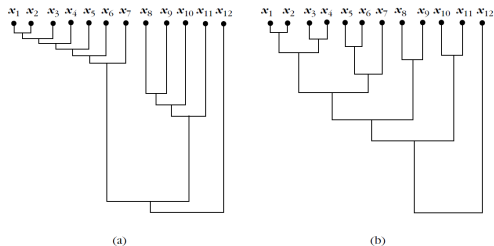


Figure: A) The dendrogram suggests that there are two major clusters in the data set.
 B) The dendrogram indicates that there is a single major cluster in the data set.

Method I

This is an **extrinsic method**, in the sense that it requires determination of the value of a **specific parameter by the user**.

It involves the definition of a function $h(C)$ that measures the **dissimilarity** between the vectors of the **same cluster** C . That is, we can view it as a **self-similarity measure**.

For example,

$$\begin{aligned} h_1(C) &= \max\{d(x, y), x, y \in C\} \\ h_2(C) &= \text{med}\{d(x, y), x, y \in C\} \end{aligned}$$

When d is a **metric distance**, $h(C)$ may be defined as

$$h_3(C) = \frac{1}{2|C|} \sum_{x \in C} \sum_{y \in C} d(x, y)$$

Method I

Let θ be an appropriate **threshold** for the adopted $h(C)$.

Then, the **algorithm terminates** at the R_t clustering if

$$\exists C_j \in R_{t+1} : h(C_j) > \theta$$

In words, R_t is the **final clustering** if there exists a cluster C in R_{t+1} with dissimilarity between its vectors greater than θ .

Method I

Sometimes the threshold θ is defined as

$$\theta = \mu + \lambda\sigma$$

where μ is the **average distance** between any two vectors in X and σ is its **variance**.

The parameter λ is a **user-defined parameter**.

Thus, the need for specifying an appropriate value for θ is transferred to the choice of λ . However, λ may be estimated more reasonably than θ .

Method II

This is an **intrinsic method**; that is, in this case only the structure of the data set X is taken into account.

According to this method, the **final clustering** R_t must satisfy the following relation:

$$d_{\min}^{ss}(C_i, C_j) > \max\{h(C_i), h(C_j)\}, \quad \forall C_i, C_j \in R_t$$

In words, in the final clustering, the **dissimilarity between every pair of clusters** is larger than the **self-similarity** of each of them (note that this is only a necessary condition).

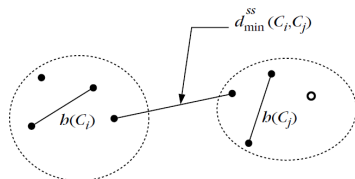


Figure: Illustration of the termination condition for method II

References

- Theodoridis, S. & Koutroumbas, K. *Pattern Recognition*, 4th ed., Academic Press, 2009